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C PROGRAMMING PROJECT PRESENTATION

Project Title: [MINI LIBRARY CATALOG]

Course: C Programming Lab / Mini Project

Student Name & Roll No: Sandhiya p [25bcs250]

Department: Computer Science / BSC.CS

Mentor / Guide: [CHRISTOPHER]

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INTRODUCTION

A library needs a proper system to maintain the details of books. In many small libraries, book records are maintained manually, which can lead to errors and difficulty in managing the data.

The Mini Library Catalog is a simple program developed using the C programming language. This system helps to manage book information in an easy way. The program allows the user to add books, issue books, return books, and display the list of books.

The system stores important details such as Book ID, Title, Author, and Status. This project helps in understanding basic concepts of C programming such as structures, arrays, loops, and functions.

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OBJECTIVE

The main objectives of this project are:

To develop a simple library management system using C.

To maintain book records efficiently.

To practice programming concepts like structures, arrays, and functions.

To allow users to add, issue, return, and view books.

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PROBLEM STATEMENT

Managing book records manually in a library can be time-consuming and may lead to mistakes. There should be a simple computerized system to manage book information.

The aim of this project is to design and implement a Mini Library Catalog system using C that allows the user to:

Add book details

Issue books

Return books

Display the list of books

This system helps to manage the library records efficiently.

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SYSTEM REQUIREMENTS

Hardware Requirements

Computer or Laptop

Minimum 2 GB RAM

Keyboard and Monitor

Software Requirements

Operating System: Windows / Linux

Programming Language: C

Compiler: Dev-C++ / GCC

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SYSTEM DESIGN

The system is designed as a menu-driven program. When the program starts, it displays a menu with different options. The user selects an option to perform an operation.

Menu Options

Add Book

Issue Book

Return Book

List Books

Exit

Each book record contains the following fields:

Book ID

Book Title

Author Name

Book Status (Available or Issued)

The program stores book information using structures and arrays.

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MODULE DESCRIPTION

The Mini Library Catalog consists of the following modules.

1. Add Book Module

This module allows the user to enter book details such as ID, title, and author name. The book is added to the catalog with status Available.

2. Issue Book Module

This module allows the user to issue a book by entering the book ID. The status of the book changes to Issued.

3. Return Book Module

This module allows the user to return an issued book. The status changes from Issued to Available.

4. List Books Module

This module displays the details of all books stored in the system.

5. Exit Module

This module terminates the program.

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ALGORITHM / FLOWCHART

Algorithm for Mini Library Catalog

Start the program

Display the menu options

Ask the user to enter a choice

If choice = 1, read book ID, title, and author and store them

If choice = 2, enter book ID and change status to issued

If choice = 3, enter book ID and change status to available

If choice = 4, display all book records

If choice = 5, exit the program

Repeat the menu until the user selects exit

Stop the program

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IMPLEMENTATION

```
#include <stdio.h>
#include <string.h>

#define MAX 100

struct Book
{
    int id;
    char title[60];
    char author[40];
    int status;
};

struct Book books[MAX];
int count = 0;
```

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```
void addBook()
{
    printf("Enter Book ID: ");
    scanf("%d",&books[count].id);
    getchar();

    printf("Enter Book Title: ");
    fgets(books[count].title,60,stdin);

    printf("Enter Author Name: ");
    fgets(books[count].author,40,stdin);

    books[count].status = 0;
    count++;

    printf("Book added successfully\n");
}
```

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```
void issueBook()
{
    int id;
    printf("Enter Book ID to issue: ");
    scanf("%d",&id);

    for(int i=0;i<count;i++)
    {
        if(books[i].id==id)
        {
            books[i].status=1;
            printf("Book issued successfully\n");
            return;
        }
    }

    printf("Book not found\n");
}
```

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```
void returnBook()
{
    int id;
    printf("Enter Book ID to return: ");
    scanf("%d",&id);

    for(int i=0;i<count;i++)
    {
        if(books[i].id==id)
        {
            books[i].status=0;
            printf("Book returned successfully\n");
            return;
        }
    }

    printf("Book not found\n");
}
```

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```
void listBooks()
{
    printf("\nID\tTitle\tAuthor\tStatus\n");

    for(int i=0;i<count;i++)
    {
        printf("%d\t%s\t%s\t%s\n",
        books[i].id,
        books[i].title,
        books[i].author,

books[i].status==0?"Available":"Issued");
    }
}

int main()
{
    int choice;

    while(1)
```

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```
{
    printf("\n--- MINI LIBRARY CATALOG ---\n");
    printf("1.Add Book\n");
    printf("2.Issue Book\n");
    printf("3.Return Book\n");
    printf("4.List Books\n");
    printf("5.Exit\n");

    printf("Enter your choice: ");
    scanf("%d",&choice);

    switch(choice)
    {
        case 1: addBook(); break;
        case 2: issueBook(); break;
        case 3: returnBook(); break;
        case 4: listBooks(); break;
        case 5: return 0;
        default: printf("Invalid choice\n");
    }
}
}
```

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OUTPUT SCREEN

```
MINI LIBRARY CATALOG ---

1.ADD BOOK
2.ISSUE BOOK
3.RETURN BOOK
4.LIST BOOKS
5.EXIT

ENTER YOUR CHOICE: 1

ENTER BOOK ID: 101
ENTER BOOK TITLE: C PROGRAMMING
ENTER AUTHOR NAME: DENNIS RITCHIE

BOOK ADDED SUCCESSFULLY
COPY CODE

ENTER YOUR CHOICE: 4

ID   TITLE          AUTHOR         STATUS
101  C PROGRAMMING    DENNIS RITCHIE   AVAILABLE
```

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RESULTS & DISCUSSION

The Mini Library Catalog system was successfully developed using C programming. The program allows users to manage book records by adding, issuing, returning, and displaying books.

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CONCLUSION & FUTURE ENHANCEMENT

This project demonstrates how C programming concepts such as structures, arrays, functions, and loops can be used to create a simple library management system.

The system can be improved in the future by adding features such as searching books, deleting books, and file storage.

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REFERENCES / ACKNOWLEDGEMENT

The C Programming Language – Kernighan and Ritchie

Online C programming tutorials

Classroom lecture notes

Acknowledgement

I would like to express my sincere thanks to my faculty members and institution for giving me the opportunity to complete this project successfully.

THANK YOU

Project Motto:

“Books Organized, Knowledge Accessible.”

Closing Remark:

“This Mini Library Catalog project highlights how technology can support efficient book management. We appreciate your time and look forward to your questions and feedback.”